Example 1:

	Take stats (A)	Not taking stats (A')	Total
Male (B)	84	145	229
Female (B')	76	134	210
Total	160	279	439 $n=$

(a) $P(\text{Male}) = P(B) = \frac{229}{439} = \underline{0.5216}$

$P(\text{not male}) = P(B') = 1 - P(B) = \underline{0.4784}$

(b) $P(\text{Take stats}) = P(A) = \frac{160}{439} = \underline{0.3645}$

(c) $P(\text{Male and take stats})$

$= P(B \cap A) = \frac{84}{439} = \underline{0.1913}$

(d) $P(\text{Male or take stats})$

$= P(B \cup A) = P(A) + P(B) - P(A \cap B)$

$= 0.3645 + 0.5216 - 0.1913 = \underline{0.6948}$

Example 2:

		Actually purchased TV		
		YES (B_1)	NO (B_2)	Tot
Plan to purchase TV	YES (A_1)	200	50	250
	NO (A_2)	100	650	750
	Tot	300	700	1000

(a) $P(A_1) + P(A_2) = \frac{250}{1000} + \frac{750}{1000} = 1$

(b) $P(A_1 \cap B_2) = \frac{50}{1000} = 0.05$

(c) $P(A_1 \cap B_1) + P(A_1 \cap B_2) = \frac{200}{1000} + \frac{50}{1000} = \frac{250}{1000}$

$= 0.25$

(d) $1 - P(B_1) = 1 - \frac{300}{1000} = \frac{700}{100} = 0.7$

(e) $P(A_1 \cup B_1) = P(A_1) + P(B_1) - P(A_1 \cap B_1)$

$= \frac{250}{1000} + \frac{300}{1000} - \frac{200}{1000} = 0.35$

(f) $P(A_1 | B_2)$

$= \frac{P(A_1 \cap B_2)}{P(B_2)} = \frac{50/1000}{700/1000} = \frac{50}{700} = 0.0714$

(g) $P(B_2 \mid B_1) = \dfrac{P(B_1 \cap B_2)}{P(B_1)} = \dfrac{0}{0.3} = 0$

(h) $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$

$= \dfrac{250}{1000} + \dfrac{750}{1000} - 0 = 1$

(i) $P(A_1 \mid B_1)\, P(B_1)$

$= \dfrac{P(A_1 \cap B_1)}{\cancel{P(B_1)}} \cancel{P(B_1)} = P(A_1 \cap B_1) = \dfrac{200}{1000} = 0.2$

(j) Are A_1 and A_2 mutually exclusive?

$P(A_1 \cap A_2) = 0$, Yes, the are mutually exclusive

(k) Are A_1 and B_1 statistically dependent?

$P(A_1 \cap B_1) \neq P(A_1) P(B_1)$.

$0.2 = \dfrac{200}{1000} \neq \dfrac{250}{1000} \times \dfrac{300}{1000} = 0.075$

Yes, the are dependent

(l) Are events A_1 and A_2 collectively exhaustive?

Yes, because $P(A_1) + P(A_2) = 1$. Thus A_1 and A_2 takes up the entire sample space.